Artificial Intelligence semester project
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Emotion Detection Using DeBERTa-v3 on the GoEmotions Dataset

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1. Introduction

Emotion detection is a Natural Language Processing (NLP) task that focuses on identifying emotions expressed in text. Unlike traditional sentiment analysis, which usually categorizes text as positive, negative, or neutral, emotion detection aims to recognize a wider range of human emotions such as joy, anger, sadness, gratitude, fear, and surprise.

The ability to automatically identify emotions from text has applications in customer feedback analysis, social media monitoring, virtual assistants, mental health support systems, and recommendation systems. Recent advances in transformer-based language models have significantly improved the performance of NLP systems by enabling models to better understand contextual relationships within text.

The goal of this project was to develop a multi-label emotion detection system capable of identifying multiple emotions from a single sentence using the GoEmotions dataset. Initially, BERT was used as the baseline model. However, performance improvements became stagnant and signs of overfitting were observed during training. To address these limitations, the project transitioned to DeBERTa-v3-base, which provided improved contextual understanding and better overall performance.

2. Project Objectives

The primary objectives of this project were:

- Develop an emotion detection system capable of recognizing multiple emotions from text.
- Fine-tune a transformer-based language model for multi-label classification.
- Explore techniques for handling class imbalance in emotion datasets.
- Evaluate model performance using standard classification metrics.
- Implement an inference system capable of predicting emotions for unseen text.

3. Dataset and Data Preparation

Dataset

This project used the GoEmotions dataset, a large-scale dataset developed by Google Research for emotion classification tasks. The dataset contains Reddit comments annotated with 28 emotion categories, including a neutral class.

Examples of emotions in the dataset include:

- Admiration
- Gratitude

- Joy
- Anger
- Fear
- Sadness
- Surprise
- Remorse
- Neutral

The dataset presents a challenging multi-label classification problem because a single sentence may contain more than one emotion simultaneously.

Data Preparation

The dataset was loaded using the Hugging Face Datasets library. Instead of preprocessing the entire dataset beforehand, a custom PyTorch Dataset class was implemented.

The custom dataset performed preprocessing dynamically whenever a sample was requested. This approach reduced memory consumption and simplified data management.

The preprocessing pipeline included:

- Tokenization using the DeBERTa tokenizer
- Padding sequences to a maximum length of 128 tokens
- Truncating longer sequences
- Creating attention masks
- Converting emotion labels into multi-hot encoded vectors

Since the task is multi-label classification, labels were represented as 28-dimensional vectors where each position corresponds to an emotion category.

4. Model Selection and Architecture

Initial BERT Experiments

The project initially used BERT as the backbone model. While BERT produced acceptable results during early experiments, validation performance remained close to a Macro F1 score of 0.40. Additional training did not result in significant improvement and signs of overfitting began to appear.

These observations motivated the search for a stronger transformer architecture.

Why DeBERTa?

DeBERTa (Decoding-enhanced BERT with Disentangled Attention) extends the original BERT architecture by separating content information and positional information during attention computation. This enables the model to learn contextual relationships more effectively.

Due to its improved language understanding capabilities, DeBERTa-v3-base was selected as the final model for this project.

Model Architecture

The final model consisted of three components:

DeBERTa-v3-base Backbone

The transformer backbone was responsible for generating contextual representations of input text.

Dropout Layer

A dropout layer with a probability of 0.3 was added after the transformer output to reduce overfitting and improve generalization.

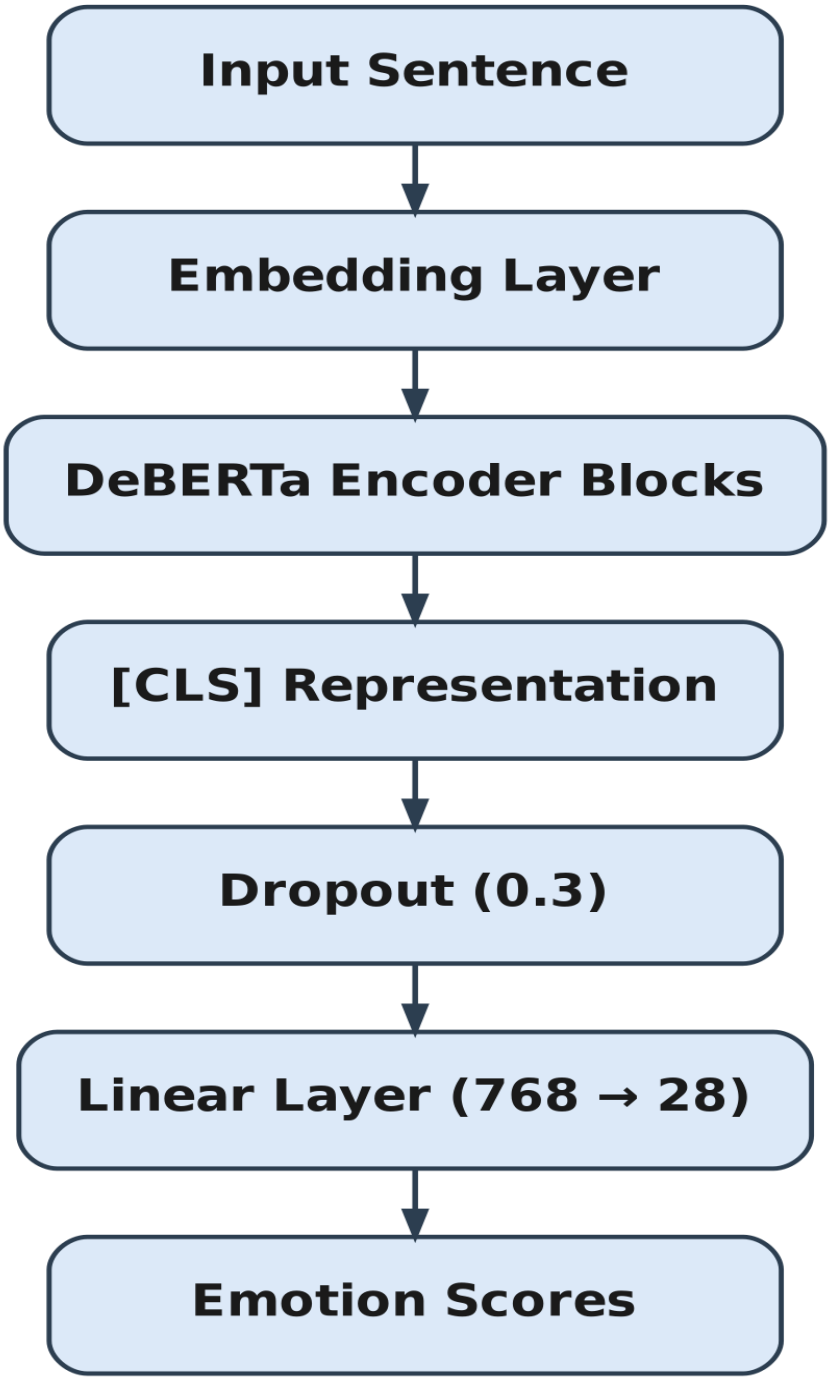
Classification Head

A fully connected linear layer mapped the 768-dimensional transformer representation to 28 emotion scores.

The complete model architecture can be summarized as:

Input Text → DeBERTa-v3-base → Dropout → Linear Layer → 28 Emotion Outputs

Unlike feature extraction approaches, the entire DeBERTa model was fine-tuned during training. All model parameters were updated, allowing the network to adapt specifically to the emotion detection task.


Input Sentence
Embedding Layer
DeBERTa Encoder Blocks
[CLS] Representation
Dropout (0.3)
Linear Layer (768 → 28)
Emotion Scores

5. Training Methodology

Training Configuration

The following hyperparameters were used during training:

Parameter	Value
Model	DeBERTa-v3-base
Batch Size	8
Gradient Accumulation	4
Effective Batch Size	32
Epochs	5
Learning Rate	1e-5
Optimizer	AdamW
Weight Decay	0.01
Dropout	0.3
Maximum Sequence Length	128

Loss Function

Because multiple emotions can be present simultaneously, Binary Cross Entropy with Logits Loss (BCEWithLogitsLoss) was used.

This loss function is suitable for multi-label classification tasks because each emotion is treated independently.

Handling Class Imbalance

The GoEmotions dataset contains significant class imbalance. Emotions such as gratitude and neutral appear frequently, while emotions like grief and relief occur much less often.

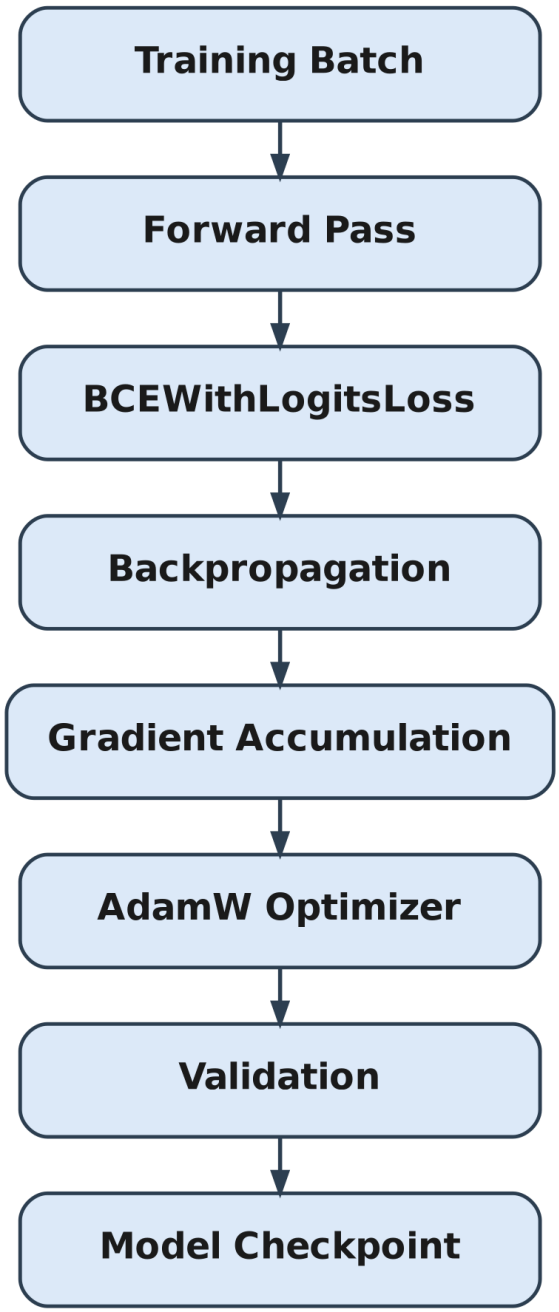
To address this issue, class-specific positive weights were calculated using a square-root dampening strategy. These weights increased the importance of rare emotions during training without causing unstable optimization.

Mixed Precision Training

Mixed precision training was implemented using PyTorch Automatic Mixed Precision (AMP). This reduced memory usage and improved training speed on GPU hardware.

Gradient Accumulation

Gradient accumulation was used to simulate a larger batch size while operating within GPU memory limitations.


Training Batch
Forward Pass
BCEWithLogitsLoss
Backpropagation
Gradient Accumulation
AdamW Optimizer
Validation
Model Checkpoint

6. Results

Evaluation was performed using the unseen test split of the GoEmotions dataset.

The model generated probabilities using the sigmoid activation function. A threshold of 0.5 was applied to determine whether a specific emotion was present.

Final Performance

Metric	Score
Micro F1	0.56
Macro F1	0.49
Weighted F1	0.58
Sample Average F1	0.59

The final training run achieved:

- Training Loss: 0.1177
- Validation Loss: 0.1873
- Validation Macro F1: 0.5110

Per-Class Performance

The model achieved strong performance on emotions that had larger numbers of training examples.

Best performing emotions included:

- Gratitude
- Amusement
- Love
- Admiration

Lower performance was observed for:

- Grief
- Nervousness
- Relief
- Realization

This behavior is expected because these emotions have relatively few examples within the dataset.

Overall, the results demonstrate that the model successfully learned meaningful emotional patterns and generalized reasonably well to unseen data.

7. Challenges and Solutions

Challenge 1: BERT Performance Plateau

Initial experiments using BERT showed limited performance improvements and validation scores remained close to 0.40 Macro F1.

Solution

The project switched to DeBERTa-v3-base, which provided stronger contextual representations and improved generalization.

Challenge 2: Class Imbalance

Several emotions appeared infrequently in the dataset, making them difficult to learn.

Solution

Class-specific positive weighting was incorporated into the loss function to increase the importance of underrepresented emotions.

Challenge 3: Training Time

Training on a local RTX 3050 Ti GPU required approximately 50 minutes per epoch.

Solution

Training was moved to Google Colab using an NVIDIA T4 GPU, reducing training time to approximately 10 minutes per epoch and significantly improving experimentation speed.

8. Additional Features

Beyond model training and evaluation, additional inference utilities were implemented.

Single Sentence Prediction

The trained model can analyze individual sentences and predict all detected emotions along with confidence scores.

Emotional Timeline Analysis

An experimental feature was developed to analyze emotional progression within longer paragraphs.

The paragraph is divided into beginning, middle, and end sections. Each section is evaluated separately, allowing visualization of how emotions change throughout a narrative.

This feature demonstrates a potential extension of the project beyond standard sentence-level classification.

9. Learning Outcomes

This project provided practical experience in several areas of Artificial Intelligence and Natural Language Processing.

Key learning outcomes included:

- Understanding transformer-based language models
- Fine-tuning pre-trained NLP models
- Multi-label classification techniques
- Handling class imbalance in real-world datasets
- GPU-based deep learning workflows
- Model evaluation and performance analysis

The project also highlighted the importance of experimentation when selecting model architectures and training strategies.

10. Future Improvements

Several improvements can be explored in future work.

- Hyperparameter optimization using systematic search methods
- Threshold tuning for individual emotions
- Emotion tracking for conversations and chat systems

These improvements may further increase classification accuracy and practical usability.

11. References

[1] Demszky, D., Movshovitz-Attias, D., Ko, J., Cowen, A., Nemade, G., and Ravi, S. "GoEmotions: A Dataset of Fine-Grained Emotions."

[2] He, P., Liu, X., Gao, J., and Chen, W. "DeBERTa: Decoding-enhanced BERT with Disentangled Attention."

[3] Devlin, J., Chang, M., Lee, K., and Toutanova, K. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding."